KOR STRUCTURAL · DRAFTING OPERATIONS

The KOR Virtual Drafter — What It Is and How We Use It

A plain-language playbook. Prepared by Ian Lalonde · July 30, 2026

What it is, in one paragraph

The Virtual Drafter is software on one KOR workstation that does drafting work in Revit the same way a careful junior drafter would: it reads the engineer's marked-up PDF, finds the affected callouts and rebar in the model, writes out a numbered change plan in plain drafting language, **waits for a human to approve it**, then makes exactly those changes and proves each one with before-and-after pictures. Engineers keep working exactly the way they always have — mark up a PDF in Bluebeam, hand it over. Nothing about how you work changes.

0
changes it can make to a live project model

100%
of changes need human approval first

1
Ctrl+Z undoes any single change it makes

Every
change logged with evidence images

What it is NOT

- **It does not design.** It has no engineering judgment and never guesses. When a markup is ambiguous, it asks a question instead of picking an answer — that behaviour is built in and was proven in testing.
- **It cannot touch live models.** It works only on disconnected copies. The software physically refuses to write into any model that is connected to a project central — even if asked.
- **It does not replace review.** Its work goes through the same eyes drafting work always has. It just arrives already checked once, with photographic evidence attached.

How a markup flows through it

1. **Engineer marks up a PDF in Bluebeam** — clouds, callouts, cross-outs. Exactly as today. Nothing new to learn.
2. **The PDF goes into a folder.** The Virtual Drafter reads the actual markup data inside the file — the real text and positions of every cloud and callout, not a guess from a picture.
3. **It studies the model** (a disconnected copy) and locates every element the markup refers to, using the sheet's own grid system to navigate.
4. **It writes a CHANGE PLAN** — a numbered list in drafting language: each markup, the element it found, current value, new value. Anything unclear is listed as a QUESTION. **It stops here and waits.**
5. **A human approves the plan** (or corrects it, or answers the questions). No approval, no changes — ever.
6. **It executes and proves it.** Each change is made, re-read from the model to confirm it took, and photographed before and after. A change report lists every markup item, what was done, and the evidence.
7. **Humans put it in production.** The verified changes are applied to the real project model through the normal drafting process.

Proof: a real revision, start to finish

On July 29 we ran a controlled test using a real markup — Jim's July 27 revision of the Hotel Circle North reinforcing plan (sheet S2.08.2, levels 6–12). The Virtual Drafter was given only the marked-up PDF and a copy of the model as it existed *before* the revision was drafted, and asked to do the drafting itself. Then we compared its work against the drafting a human had already done for that same markup.

Measure	Result
Markup locations identified and executed	**12 of 12, on both affected sheets (levels 6–12)**

Callout notes rewritten, bars re-parameterised, new bars placed & tagged	**38 edits, 10 placements, 2 deletions — every one verified by re-reading the model**
Agreement with the human drafter's version	**Matched or equivalent at 10 of 12 locations**
The other 2 locations	**The machine followed the engineer's markup exactly; the human version differed from the markup (12 vs 14 bars at one grid; #5 vs #6 at another). Both are now questions back to the engineer — the test caught two discrepancies in issued work**
Ambiguities guessed at	**Zero — unclear items became written questions**

The behaviour that matters most: midway through testing, it found it could not uniquely identify which of 13 identical bars a markup pointed at. It stopped and said so: "*picking one of 13 by guess is the wrong-bar-in-concrete case, so I stopped.*" That is the discipline we want from every junior — human or not.

What it knows (and how it learned it)

Overnight on July 29–30, it read **every structural Revit model KOR has** — 198 models across ten years of Revit versions — without changing a single one. From that it built a reference library: which rebar families and callout conventions each job uses, how each project's sheets are organised, and where the data has problems. That reading pass alone surfaced findings we are already acting on:

- **~4,000 bars across the fleet print "NA" instead of a length** on their tags — a gap in a hand-typed lookup table inside the tag family. Fix in progress (the tag will calculate the value instead).
- **Two corrupt salvage copies** of one tower model flagged for cleanup; link problems catalogued per job.
- Per-job records of drafting conventions, so future work starts already knowing each job's habits.

Standards: three reviews, finally one standard

Our CAD and Revit standard sheets drifted apart over the years, and three people reviewed the problem independently — Simon's line-by-line comparison in May, Kevin's engineering review in June, Rory's unit rulings — but the three reviews were never combined. They now are. Each reviewer's recommendations are accepted as given; the small number of genuinely open questions has gone back to each person **inside their own marked-up PDF**, at the exact spot, answerable in Bluebeam. When the answers return, both the CAD and Revit master sheets get corrected in one pass, verified until they match exactly, and re-checked automatically from then on so they can never silently drift again. The Virtual Drafter then drafts to that ratified standard on every job it touches.

Who does what now

People	They keep / gain
Engineers	Mark up PDFs exactly as always. Rule on standards questions when asked — one word in a box. Gain: revisions come back faster, with evidence, and discrepancies get caught instead of buried.
Drafters	All judgment work: complex details, sheet composition, grouped edits (some Revit operations only a human can perform — that is a hard limit of Revit itself), review of the machine's output. Gain: the repetitive transcription work — retyping a markup into the model — stops being their job.
The Virtual Drafter	Mechanical transcription of markups, verification, evidence, fleet-wide audits, standards enforcement. The work nobody misses.

The safety system, in plain words

- **Copies only.** It works on disconnected copies of models. The live project files are physically untouchable by it — the software refuses, by design, and this was tested.
- **Approval gate.** No model change of any kind before a human approves the written plan.
- **All-or-nothing edits.** A batch of changes either fully succeeds or fully cancels itself — it can never half-apply and leave a mess.
- **Everything is evidence.** Every command, change, and even every dialog box it dismissed is logged; every changed area is photographed before and after.
- **Errors stop the machine.** A real error never gets auto-answered — it waits for a person. In testing it disclosed its own mistakes unprompted, including three misplaced items it caught and removed itself.

Where this goes next

- **Live pilot:** the first production task is applying the engineer's confirmed corrections from the Hotel Circle test back to the real project — small, checked, real.
- **Standards execution:** when the standards answers come back, the master sheets get fixed in one pass.
- **Quantity data:** the same reading ability is producing per-job rebar inventories that feed estimating (bar counts today, weight-grade takeoff data next).